Maximum flow: exercises

26-E01 Antiparallel arc flow, net flow, and residuals

Capacities are s→a=4, s→b=1, a→b=3, b→a=2, a→t=1, b→t=4. Given nonnegative arc flows g_sa=3, g_sb=1, g_ab=3, g_ba=1, g_at=1, g_bt=3, check conservation, convert to net flow, and find residual capacities a→b, b→a. Augment once and give an equal cut proving final optimality. Do not impose directionwise nonnegativity on antisymmetric net flow.

26-E02 Flow and cut certify opposite bounds

Capacities are s→a=5, s→b=3, a→b=2, a→t=3, b→t=4. Construct a maximum flow, check conservation and capacities, and give an equal cut. Do source outcapacity8 and sink incapacity7 alone prove maximum7? Identify the missing evidence.

26-E03 An actual pseudopolynomial path family

For positive integer C, let s→a=s→b=a→t=b→t=C, a→b=1. Alternate paths s→a→b→t and s→b→a→t. Give all five net flows after k pairs and prove exactly 2C augmentations occur. For C=2^B, relate this to input bit length. How many BFS augmentations suffice on the same graph?

26-E04 Matching flips and a Hall obstruction

Edges are L1R1, L1R2, L2R1, L3R2, L3R3. Starting from{L1R1, L3R2}, find an alternating augmentation, give the final maximum matching and an equal-size vertex cover. Delete L3R3, find a left set violating Hall, and prove full left coverage impossible.

26-E05 No residual s-t path can still be unfinished

For s, u, t with s→u=2, u→t=1, initialize push-relabel and list f, e, h, residual arcs. Is there an s-to-t residual path? Why does max-flow min-cut not yet prove maximum flow? Complete the operations, including return of excess to s.

26-E06 Neighbor order changes distribution, not optimal value

Use the main five-vertex network with N(a)=[t, b, s], N(b)=[t, c, a, s], N(c)=[t, b, s], initial L=[a, b, c]. List every push amount and excess, then list L, current, and heights after each discharge. Give every final original edge flow and a minimum cut. Must b→c, b→t, c→t match the main trace?

Maximum flow: exercises (continued)

26-E07 Safety and cost of current pointers

With fixed h(u), arc uv is skipped because residual is 0. If other operations restore its residual, why need current not immediately return? Also prove a formerly positive but inadmissible residual arc cannot become admissible on its own. Derive O(nm) total current advances. Which charging step fails if current resets at the start of every discharge?

26-E08 Rational capacities and exact certificates

Capacities are s→a=1/2, a→t=1/3, s→b=b→t=1/6. Give a maximum flow and minimum cut. Scale by 6, solve the integer instance, and restore the answer. Explain how a denominator product can be exponentially large in value yet polynomial in bit length, and compare Ford–Fulkerson and Edmonds–Karp round-count dependencies.

26-E09 Complete residual networks, not just paths

All original capacities s→a, s→b, a→b, a→t, b→t are 1. Augment first along s→a→b→t, then s→b→a→t, each by 1. List every positive residual arc and every original flow after each round. Prove optimality with a feasible flow and an equal cut. Why can residual a→b remain positive without further augmentation? Then change the capacities, in that order, to (3,2,1,2,3). With BFS visiting a before b, list every round's full residual network, distances d(a,b,t), path, and bottleneck through termination.

26-E10 Escape without shared cells

Only cells {(3,2),(2,3),(3,3),(3,4),(3,5),(4,3),(5,3)} in a 5×5 grid are open. Coordinates are (row,column). People start at (3,2) and (2,3); open boundary cells are exits. Find the maximum escaping count for edge-disjoint routes and for vertex-disjoint routes. Supply actual routes and matching cuts. For a general k×k grid, construct the split network, bound its size, prove both directions of the integral-flow correspondence, and handle k=1 and boundary starts.

26-E11 Reverse rerouting and a six-vertex cut certificate

Capacities are sw:8, wx:9, xy:3, xt:4, sy:10, yz:7, zt:8. Augment in order along s-w-x-y-z-t, s-y-x-t, s-y-z-t, s-w-x-t. For each round list the path residuals, bottleneck, seven original flows, and total value. Explain when y→x becomes available. Finally give the residual-reachable set, every original cut arc, and an optimality proof. Why is s-y-x-t invalid as the initial path?

Maximum flow: exercises (continued)

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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